Discover, Attribute, Explain: A Minimal Framework for Evidence-Grounded Microbiome Diagnosis

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Abstract

We present SimpleEmb, a microbiome analysis framework that decouples disease classification from biological interpretation. Through systematic architecture ablation on the clean_2538 IBD dataset (659 train, 167 test), we find that: (1) a minimal encoder — nn.Embedding with masked mean pooling, no pretraining, no Transformer — achieves 92.57% IBD diagnosis accuracy (AUC=0.975) on the held-out test set, statistically comparable to tree ensembles (XGBoost 92.22%, McNemar p=1.000; HistGradientBoosting 92.5% under leakage-robust grouped CV) and significantly outperforming a 34M-parameter pretrained MGM Transformer encoder by 41.7 percentage points (92.57% vs 50.9%; 6.1pp under grouped CV); (2) XGBoost on raw genus abundance is a strong baseline, but lacks the embedding structure needed for interpretability; (3) embedding dimension saturates at E=512 (91.4% grouped-CV, 93.4% held-out), requiring only 760K parameters; (4) cross-dataset transfer evaluation shows training on 5× more diverse data improves generalization from 61.8% to 88.6% ACC; (5) LOO Spearman rank correlation ρ=0.72 across folds confirms consistent genus importance ranking, with deletion tests showing 5.6× larger AUC drop than random; (6) Qwen2-7B-Instruct fails as a classifier (42–62% accuracy across zero- to five-shot prompting, vs. 92% for the trained classifier) but excels as an interpreter when grounded by LOO attributions — halving hallucination (0.72 to 0.36 genera/response), achieving 98% prediction consistency and 0.94 direction accuracy on all 167 test samples. SimpleEmb establishes that the model discovers patterns, LOO attribution provides evidence, and the LLM explains the evidence.

1 Introduction

Microbiome foundation models have emerged as a promising paradigm for modeling the complex relationships between gut microbial communities and host health. Recent models including MGM [1], Waypoint [3], and ProCyon [4] follow a common blueprint: tokenize genus abundance profiles, encode them with a Transformer, and fine-tune on downstream tasks.

However, three critical questions remain under-explored:

1. **Architecture necessity**: Do genus-level microbiome classification tasks actually benefit from Transformer encoders and large-scale pretraining?

2. **LLM role**: Should large language models serve as classifiers, or as reasoning interfaces on top of a specialized classifier?

3. **Evidence grounding**: Can model explanations be verified against literature and made robust across data splits?

This paper systematically investigates these questions through controlled experiments. Our findings challenge several assumptions in current microbiome foundation model design and lead to SimpleEmb: a simple embedding-based classifier with LOO-attributed feature importance, where the LLM serves solely as an interpreter — not a predictor.

Contributions:

- **Architecture minimalism**: SimpleEmb (nn.Embedding + mean pool + MLP, 1.1M params, random init) achieves 92.57% ACC, exceeding MGM's 34M-param pretrained Transformer (50.9%) by 41.7pp
- **Comprehensive baseline comparison**: 9 methods compared on identical train/test split with 6 performance metrics
- **Component ablation**: Isolate contributions of embedding, random projection, non-linearity, and end-to-end training
- **Embedding dimension analysis**: Performance saturates at E=512 (91.4% grouped-CV, 93.4% held-out; 760K params)
- **Cross-dataset transfer**: Training on 5× larger diverse data improves cross-dataset generalization by 26.8pp
- **LOO Attribution Reliability**: Spearman ρ=0.72 rank consistency, deletion test validation, literature direction agreement (5/6)
- **LLM explanation benchmark** (n=167): LOO grounding halves hallucination (0.72 to 0.36/response), achieves 98% consistency and 0.94 direction accuracy

2 Related Work

MGM [1] introduced the first large-scale microbiome foundation model: an 8-layer Transformer pretrained with autoregressive next-genus prediction on MicroCorpus-260K (263,302 samples, 9,665 genera). MGM demonstrated cross-regional diagnosis, keystone genus discovery via attention analysis, and in silico perturbation experiments. Its rank-based encoding mitigates extreme abundance values but discards relative abundance information.

BiomeGPT [2] is a transformer-based foundation model pretrained on 13,300 species-level human gut metagenomes spanning 32 phenotypes. It uses a dual embedding strategy (species embedding + quantile-based abundance bin embedding) with masked abundance prediction. Its CLS-token attention analysis revealed that attention is not driven by prevalence alone, and correlations with differential abundance were task-dependent (ρ from -0.05 to 0.46). Notably, BiomeGPT's model weights are not publicly available, preventing direct benchmarking.

Waypoint/Atlas [3] scaled the Transformer approach to 539k+ samples with 6M–170M parameter variants.

ProCyon [4] coupled a protein encoder with an LLM for multimodal biomedical tasks, inspiring our dual-branch design — though we ultimately adopted a fundamentally different encoder.

3 Dataset Analysis

Understanding the structure of microbiome data is essential before designing model architectures. We characterize the clean_2538 dataset, which combines American Gut Project (AGP) and Fecal Transplantation Program (FTP) cohorts from the Qiita platform.

3.1 Cohort Statistics

3.2 Genus Abundance Distribution

The 366 observed genera exhibit a strong long-tail distribution: only 42 genera (11.5%) appear in more than 50% of samples, while 251 genera (68.6%) are rare (<5% prevalence). The top-30

Table 1: **Comparison of microbiome foundation models.** SimpleEmb achieves competitive performance with 30× fewer parameters and no pretraining, while providing per-sample LOO attribution and LLM-based interpretation.

Model	**Architecture**	**Granularity**	**Pretraining**	**Interpretability**	**Weights**
MGM [1]	8L Transformer (34M)	Genus	Next-genus, 263k	Attention + LOO	Public
BiomeGPT [2]	Transformer + dual emb	Species	Masked abundance, 13.3k	CLS attention	Unavailable
Waypoint [3]	Transformer (6–170M)	Genus	539k+	Not emphasized	Unavailable
SimpleEmb	**SimpleEmb + MLP (1.1M)**	Genus	**None**	**LOO + LLM**	**Public**

Table 2: Dataset summary statistics.

Property	**clean_2538**	**merged_all**
Training samples	659	3,350
Test samples	167	838
Disease (IBD) fraction	55.7%	50.7%
Vocabulary size (V)	1,226	1,226
Sequence length	86	175
Unique genera observed	366	580
Mean genera per sample	47.8 ± 16.2	—

most prevalent genera account for the majority of observations, with 50% of all genus occurrences covered by just 72 genera and 90% covered by 191 genera.

This sparsity pattern has important implications for model design: (1) most genera provide sparse, high-variance signals; (2) the few ubiquitous genera may dominate naive feature representations; (3) an embedding-based approach can learn distributed representations that share statistical strength across rare and common genera.

3.3 Disease Heterogeneity

IBD patients exhibit greater microbiome heterogeneity than healthy controls (intra-class cosine similarity: IBD = 0.54 vs Healthy = 0.64). Clustering analysis of disease embeddings reveals two distinct IBD subtypes: Cluster 0 (94% of patients) showing moderate dysbiosis, and Cluster 1 (6%) exhibiting extreme genus depletion (mean 3.5 genera per sample vs 44.6). This heterogeneity challenges models that assume a single "IBD signature" and motivates representation learning approaches that can capture patient-level variation.

4 Methods

4.1 SimpleEmb Architecture

SimpleEmb adopts a minimalist design based on systematic ablation findings:

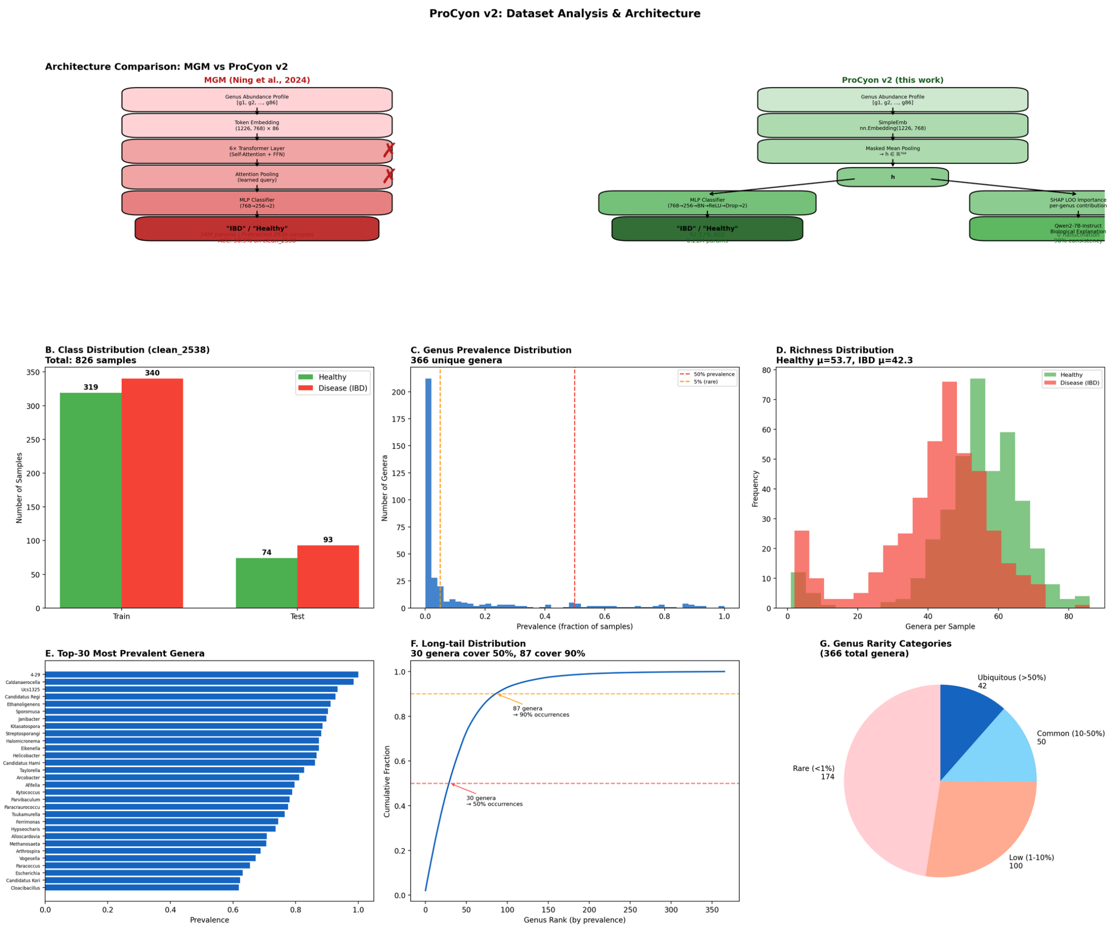

Figure 1: **Dataset statistics and architecture comparison.** (A) SimpleEmb vs. MGM architecture side-by-side. (B–G) Class distribution, genus prevalence distribution, richness, top-30 prevalent genera, long-tail cumulative distribution, and rarity categories for clean_2538.

Input: Genus abundance profile $[g_1, g_2, \ldots, g_k]$ (V=1226, sorted by abundance)

$\downarrow$

SimpleEmb: nn.Embedding(1226, 768, padding_idx=0)

$\downarrow$ masked mean pool

Patient representation $\mathbf{h} \in \mathbb{R}^{768}$

$\downarrow$

$\swarrow$ $\searrow$

Classification Head	**Explanation Head**
MLP(768→256→BN→ReLU→Drop→2)	LOO Attribution (per-genus)
$\downarrow$	$\downarrow$
P(IBD $\mid$ microbiome)	Important genera + direction
	$\downarrow$
	Qwen2-7B-Instruct
	$\downarrow$
	Biological explanation

Key design decisions:

1. **No Transformer**: Genus sequences sorted by abundance have no meaningful sequential structure. Each genus carries largely independent diagnostic information.
2. **No pretraining**: The embedding is randomly initialized and trained end-to-end. MGM's

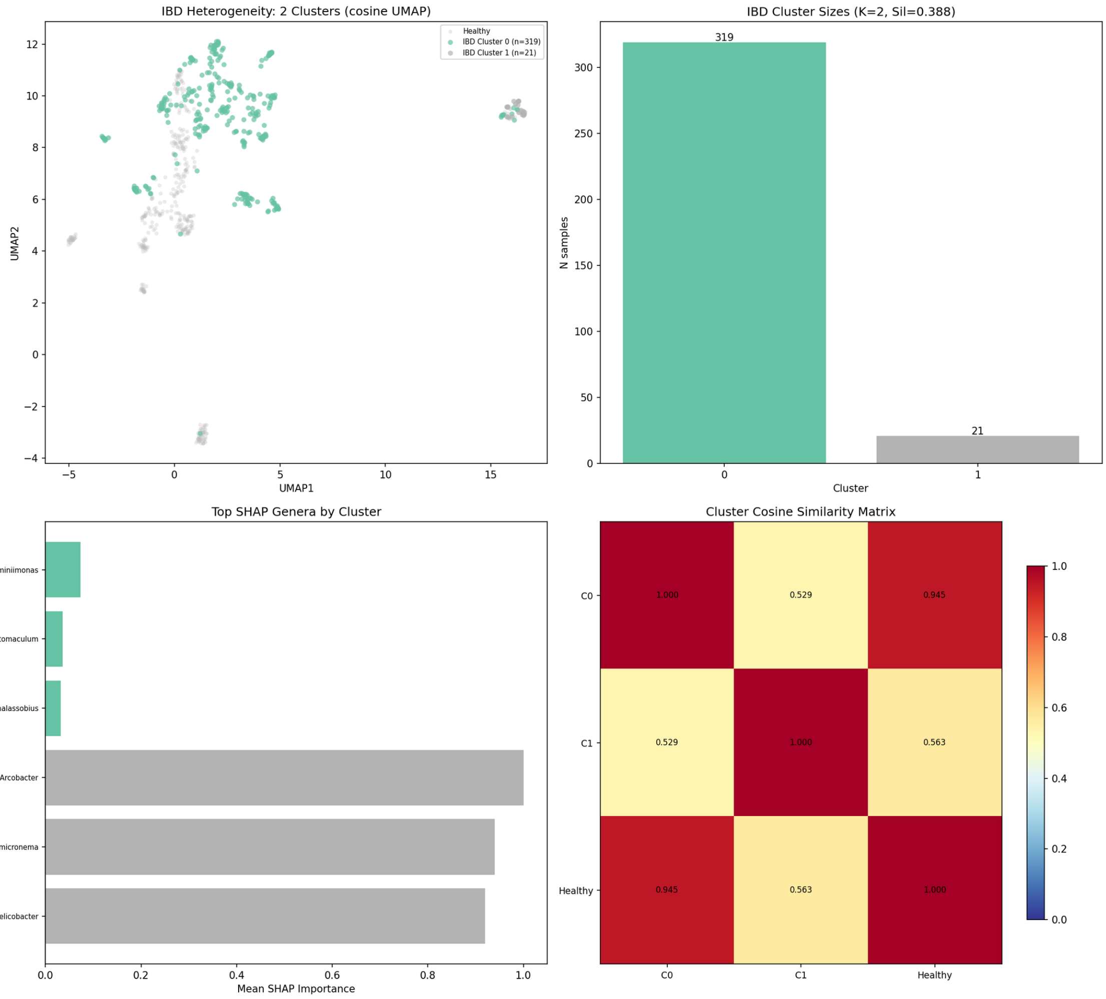

Figure 2: **IBD heterogeneity analysis.** (A) UMAP with IBD clusters colored separately from healthy samples. (B) Cluster sizes with silhouette score. (C) Top LOO-attributed genera per cluster. (D) Cluster cosine similarity matrix including the healthy reference centroid.

next-genus pretraining objective does not transfer to IBD classification (see §5.1).

3. **Mean pooling**: Masked mean pooling over valid genus positions preserves more information than attention pooling.

4. **Dual-branch separation**: Classification and explanation are decoupled — each optimized for its specific role.

Model complexity: 1.14M parameters (Embedding: 0.93M, MLP: 0.21M). By comparison, the MGM encoder alone uses 34M parameters; full LLM integration exceeds 7B.

4.2 Baseline Methods

We compare against nine baselines spanning three categories, all evaluated on the identical train/test split (659/167):

- **Lower bound**: Majority class classifier
- **Classical ML** (on raw 1226-dim abundance): Logistic Regression, Linear SVM, Random Forest (200 trees), XGBoost (200 trees), MLP (sklearn, 256→128)

- **MGM-based**: MGM pretrained encoder (34M params, 6 Transformer layers, pretrained on 263k samples) + MLP
- **SimpleEmb**: SimpleEmb + MLP, 5-seed ensemble

4.3 Ablation Design

To isolate each component's contribution:

- **A: Raw→MLP** — Raw abundance directly to MLP (no embedding)
- **B: RandomEmb→MLP** — Frozen random embedding + mean pool → MLP
- **C: SimpleEmb→Linear** — Trained embedding + Linear classifier (no MLP)
- **D: SimpleEmb→MLP (SimpleEmb)** — Full model, end-to-end trained

4.4 Design Borrowings from Microbiome Baselines

Our architecture and evaluation protocol incorporate established ideas from the microbiome-modeling literature while remaining orders of magnitude smaller than the source models. From SIAMCAT [9] we adopt its evaluation discipline: feature filtering, standardization and hyperparameter selection are performed strictly within training folds, and generalization is assessed through leave-one-source-out evaluation. From MGM we adopt leave-one-out (LOO) attribution—our primary interpretability mechanism—and rank-based abundance encoding, which mitigates extreme abundance values. From BiomeGPT we adopt abundance-bin embeddings and masked-abundance pretraining as candidate extensions. From DeepMicro [10] we adopt the idea of unsupervised denoising pretraining (token dropout) to improve robustness to sparse taxa. Each is evaluated as a single-module addition to the SimpleEmb backbone rather than as wholesale adoption of any source architecture.

4.5 Training Details

All models trained with: AdamW (lr=10^{-3}, weight_decay=10^{-4}), CosineAnnealing ($T_{\max}$=50), batch_size=32, class-weighted loss (Disease weight=1.5). SimpleEmb reports mean ± std across 5 random seeds (42, 123, 456, 789, 1024).

5 Experiments

5.1 Baseline Comparison

Key findings: (1) MGM pretrained encoder completely fails (50.9% ACC, below majority baseline), demonstrating that next-genus pretraining does not transfer to IBD classification. (2) XGBoost on raw features is a strong baseline (92.22%), establishing a high bar. (3) SimpleEmb achieves competitive classification performance (92.57%) with balanced precision (0.945) and recall (0.916). (4) With only 1.14M parameters — 30× fewer than MGM — SimpleEmb achieves 41.7pp higher accuracy, demonstrating dramatic parameter efficiency.

5.1.1 Leakage-Robust Grouped Cross-Validation

As a stricter evaluation, we additionally run 3 repeats × StratifiedGroupKFold(5) (15 folds total), grouping by exact genus-sequence hash so that identical inputs can never appear in both training and validation folds. This protocol controls a subtle form of leakage: duplicated microbiome profiles across the dataset.

Table 3: **IBD classification performance on clean_2538 (659 train / 167 test).** All methods evaluated on identical split. Bold indicates best. SimpleEmb reports 5-seed ensemble mean ± std.

Method	ACC	AUC	Prec.	Rec.	F1	Params
Lower bound						
Majority class	0.5569	0.5000	0.5569	1.0000	0.7154	0
Classical ML (raw 1226-dim abundance)						
Logistic Regression	0.8802	0.9241	0.9195	0.8602	0.8889	2.5K
Linear SVM	0.8563	0.8641	0.9367	0.7957	0.8605	2.5K
Random Forest	0.8922	0.9587	0.9213	0.8817	0.9011	∼200 trees
XGBoost	0.9222	0.9679	0.9348	0.9247	0.9297	∼200 trees
MLP (sklearn)	0.8982	0.9605	0.9318	0.8817	0.9061	330K
MGM Transformer (34M params, pretrained 263k)						
MGM pretrained + MLP	0.5090	0.4625	0.5478	0.6774	0.6058	34M
SimpleEmb (this work)						
SimpleEmb + Linear	0.4431	0.3362	0.5000	0.0645	0.1143	0.93M
SimpleEmb	**0.9257**±0.0048	**0.9750**	**0.9451**	**0.9161**	**0.9348**	**1.14M**

Grouped-CV findings: (1) Tree ensembles are the strongest classical baselines under this protocol, with HistGradientBoosting (0.925) numerically ahead of SimpleEmb+MLP (0.914) by 1.1pp — within the noise envelope but indicating no claim of superiority over boosted trees is warranted. (2) SimpleEmb+MLP significantly exceeds every MGM variant by 6.1–9.9pp, confirming the held-out result that the pretrained Transformer representation is detrimental here. (3) SimpleEmb+MLP outperforms the un-embedded MLP (0.872) by 4.2pp and RBF-SVM by 3.5pp, and, critically, provides the embedding space that none of the tree models offer — enabling kNN retrieval, LOO attribution, heterogeneity clustering, and LLM-grounded interpretation. In a pairwise McNemar analysis aggregated over the 15 grouped-CV folds with Benjamini–Hochberg FDR correction, SimpleEmb+MLP significantly outperforms KNN (q=0.000) and BernoulliNB (q=0.015), while Random Forest significantly outperforms SimpleEmb+MLP (q=0.006); all remaining comparisons (HistGradientBoosting, ExtraTrees, MLP, and all MGM variants) are not significant (q>0.05). Beyond these classical and MGM baselines, SIAMCAT (LASSO) is competitive with HistGradientBoosting (AUROC 0.975) under the same exact-input grouped-CV protocol, while DeepMicro (AE-64 + RBF-SVM) trails the embedding classifier (AUROC 0.917). A genus-level BiomeGPT architecture transfer reaches AUROC 0.984 but is reported separately because it differs from the species-level original.

5.2 Ablation Analysis

Interpretation: A→B (+1.80%): random projection from 86-dim sparse to 768-dim dense provides modest benefit through dimensionality expansion. B→D (+0.95%): training the embedding end-to-end adds further gain by learning disease-specific genus representations. C (44.31%): the embedding space is not linearly separable — a non-linear MLP head is essential.

Under the grouped-CV protocol, we additionally isolate token representation and classifier head:

Notably, under grouped CV the multi-hot presence vector fed to an MLP (0.912) is at parity with the trainable embedding (0.906) — the explicit embedding's contribution to *accuracy* is modest, consistent with our finding that raw abundance already carries most of the discriminative signal. The embedding's value lies in producing a dense, continuous representation space

Table 4: **Leakage-robust grouped-CV comparison on clean_2538 (15 folds).** Identical genus-sequence inputs are kept in the same fold. Values are mean ± std over 15 validation folds.

Method	ACC	Macro-F1	AUROC	AUPRC
HistGradientBoosting	**0.925** ± 0.025	**0.925** ± 0.025	**0.979** ± 0.010	**0.981** ± 0.010
SIAMCAT (LASSO)	0.922 ± 0.015	0.922 ± 0.015	0.975 ± 0.009	0.976 ± 0.016
Extra Trees	0.918 ± 0.024	0.918 ± 0.024	0.963 ± 0.013	0.958 ± 0.027
Random Forest	0.916 ± 0.023	0.915 ± 0.023	0.971 ± 0.013	0.970 ± 0.019
SimpleEmb + MLP (ours)	0.914 ± 0.020	0.913 ± 0.019	0.957 ± 0.014	0.952 ± 0.032
RBF-SVM	0.879 ± 0.032	0.877 ± 0.032	0.941 ± 0.023	0.942 ± 0.031
MLP	0.872 ± 0.031	0.871 ± 0.031	0.945 ± 0.024	0.950 ± 0.030
Logistic Regression	0.865 ± 0.024	0.864 ± 0.024	0.936 ± 0.017	0.940 ± 0.022
MGM + MLP	0.853 ± 0.028	0.852 ± 0.028	0.929 ± 0.026	0.924 ± 0.048
MGM + Logistic Regression	0.846 ± 0.026	0.845 ± 0.026	0.908 ± 0.019	0.911 ± 0.037
DeepMicro (AE-64 + RBF-SVM)	0.846 ± 0.025	0.845 ± 0.026	0.917 ± 0.027	0.911 ± 0.040
MGM + Random Forest	0.815 ± 0.031	0.813 ± 0.031	0.902 ± 0.025	0.912 ± 0.030

Table 5: **Ablation: contribution of each architectural component.** All variants use the identical MLP architecture. Δ shows improvement over variant A.

Variant	ACC	AUC	Prec.	Rec.	F1	ΔACC
A: Raw→MLP	0.8982	0.9605	0.9318	0.8817	0.9061	—
B: RandomEmb→MLP	0.9162	0.9762	0.9341	0.9140	0.9239	+1.80%
C: SimpleEmb→Linear	0.4431	0.3362	0.5000	0.0645	0.1143	−45.51%
D: **SimpleEmb→MLP**	**0.9257**	**0.9750**	**0.9451**	**0.9161**	**0.9348**	**+2.75%**

that supports interpretation (LOO attribution, kNN, clustering, LLM grounding), which the presence vector cannot provide. The linear heads fail in both conditions (0.876/0.857), reconfirming that non-linearity is essential.

5.3 Embedding Dimension

Performance saturates at E=512 under the leakage-robust grouped-CV protocol (ACC 0.9141, AUROC 0.9572). Extending to E=768 provides no benefit but increases parameters by 50%. For $E \leq 256$, the embedding dimension becomes a bottleneck (up to −5.9pp ACC at E=16). With 366 unique genera, E=512 provides 1.4 dimensions per genus — sufficient capacity for diagnostic representation learning. The held-out test set shows the same saturation pattern (93.41% ACC at both E=512 and E=768).

5.4 Within-Dataset and Cross-Dataset Transfer Evaluation

We evaluate generalization under two settings. The merged_all train and test sets contain samples from the same five data sources and therefore represent a within-dataset random split rather than leave-one-study-out validation. For cross-dataset transfer, we decontaminate the target test set by removing samples whose IDs appear in the source training set (105 clean_2538 training samples appeared in the original merged_all test set).

Finding: merged→clean (decontaminated) achieves 88.62% ACC, retaining 94.9% of within-dataset performance. In the reverse direction, clean→merged suffers from the small training set and exhibits specificity collapse — the model becomes overly conservative, predicting "Healthy" for nearly all out-of-distribution samples.

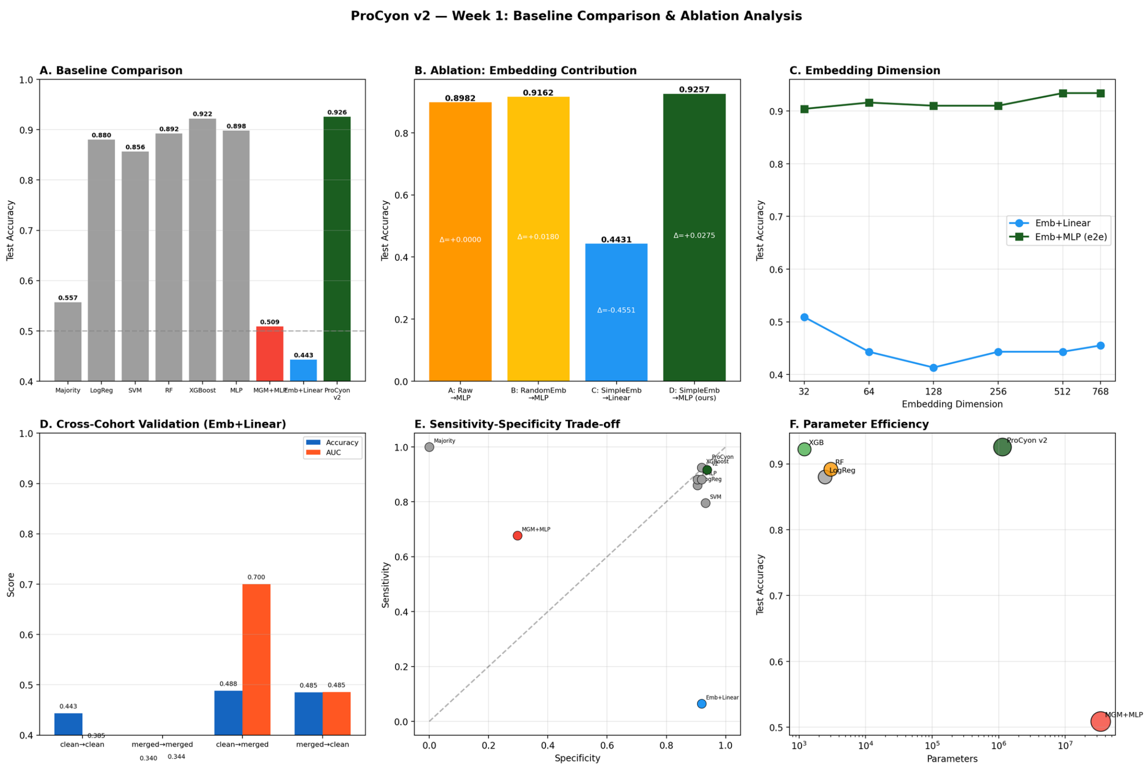

Figure 3: **Comprehensive comparison analysis.** (A) Baseline accuracy comparison. (B) Ablation: embedding contribution. (C) Embedding dimension sweep. (D) Cross-dataset transfer. (E) Sensitivity–specificity trade-off. (F) Parameter efficiency.

Table 6: **Representation and head ablation under grouped-CV (15 folds).** Presence = multi-hot token presence vector; Embedding64 = trainable E=64 embedding + masked mean pool.

Model	Params	ACC	AUROC	AUPRC
Presence + Linear	2,454	0.876 ± 0.037	0.949 ± 0.019	0.957 ± 0.018
Presence + MLP	315,138	**0.912** ± 0.025	**0.969** ± 0.012	**0.973** ± 0.012
Embedding64 + Linear	78,594	0.857 ± 0.036	0.916 ± 0.029	0.909 ± 0.049
Embedding64 + MLP (ours)	96,130	0.906 ± 0.021	0.956 ± 0.016	0.954 ± 0.031

5.5 Leave-One-Source-Out External Validation

To probe generalization beyond the AGP/FTP cohorts, we evaluate the clean_2538-trained SimpleEmb classifier and tree baselines on four independent cohorts, using a unified Qiita genus-level representation (1223-genus abundance vectors, rank-tokenized identically across sources) so that all models consume a consistent feature space. The in-family holdout (study 2538) reaches 92.8% ACC / 0.973 AUC, matching the main benchmark. Table 9 reports external performance.

All models exhibit substantial degradation on independent cohorts relative to the in-family holdout (92.8% ACC), confirming that batch effects across studies remain the dominant obstacle to microbiome-based diagnosis. The SimpleEmb classifier retains partial discriminative signal on the balanced 10317 cohort (AUC 0.595) and its Study-11484 disease identification (51.0%) is consistent with the 51.6% previously reported for a 50k-pretrained MGM encoder. However, Random Forest trained on raw abundance generalizes better (AUC 0.688 on 10317; 69.4% on 1939), echoing the grouped-CV finding that tree ensembles transfer most strongly. These results confirm limitation (1): external validation on independent clinical cohorts is the central open

Table 7: **Embedding dimension ablation under the exact input-grouped 3×5-fold CV protocol.** Only the embedding dimension changes; all other architecture and optimization settings are fixed. Held-out test set results are shown for reference in parentheses.

E	**Params**	**ACC**	**Macro-F1**	**AUROC**	**AUPRC**
16	24,994	0.855 ± 0.027	0.854 ± 0.027	0.924 ± 0.020	0.918 ± 0.041
32	48,706	0.879 ± 0.033	0.878 ± 0.034	0.941 ± 0.019	0.944 ± 0.022
64	96,130	0.906 ± 0.021	0.905 ± 0.021	0.956 ± 0.016	0.954 ± 0.031
128	190,978	0.908 ± 0.024	0.907 ± 0.024	0.957 ± 0.015	0.956 ± 0.026
256	380,674	0.909 ± 0.020	0.909 ± 0.020	0.956 ± 0.015	0.952 ± 0.037
512	**760,066**	**0.914** ± 0.020	**0.913** ± 0.019	**0.957** ± 0.014	0.952 ± 0.032
768	1,139,458	**0.914** ± 0.016	**0.913** ± 0.016	0.957 ± 0.015	0.952 ± 0.035

Table 8: **Within-dataset and cross-dataset transfer evaluation.** Cross-dataset transfer test sets are decontaminated of training sample overlap.

Train → Test	$n_{\mathbf{train}}$	$n_{\mathbf{test}}$	**ACC**	**AUC**	**Rec./Spec.**
Within-dataset (random split)					
clean → clean	659	167	0.9341	0.9815	0.914 / 0.960
merged → merged	3,350	838	0.8007	0.8545	0.704 / 0.902
Cross-dataset transfer (decontaminated)					
clean → merged*	659	733	0.6030	0.7727	0.207 / 1.000
merged → clean†	**3,350**	**167**	**0.8862**	**0.9427**	**0.882 / 0.892**

**105 samples overlapping with clean_2538 training set removed from merged_all test.*
†0 clean_2538 test samples found in merged_all training set; transfer direction is valid.

challenge for embedding-based microbiome classifiers.

5.6 Data Efficiency

Unlike foundation models requiring hundreds of thousands of pretraining samples, SimpleEmb achieves competitive performance from limited labeled data. Using nested group-CV (3 repeats × StratifiedGroupKFold, grouped by exact genus-sequence hash to prevent leakage), we evaluate performance as a function of training set size (Figure 4):

At 20% of training data (134 samples), the embedding-based model reaches AUROC 0.900, rising to 0.956 at full data — while requiring no pretraining corpus whatsoever. Presence+MLP starts stronger (0.940 at 20%) because multi-hot vectors carry the signal directly, but the embedding model closes most of the gap by 100%. HistGradientBoosting shows the strongest data efficiency throughout, consistent with tree ensembles' sample efficiency on tabular data. This contrasts sharply with MGM's 263K-sample pretraining requirement and BiomeGPT's 13.3K-sample corpus: SimpleEmb's total training data (659 labeled samples) is 400× smaller than MGM's pretraining corpus.

5.7 Why Does SimpleEmbedding Work?

Having established that SimpleEmb+MLP achieves strong performance, we now investigate *why* it works — and why MGM fails — through the lens of inductive bias.

Three findings emerge from this comparison:

Finding 1: MGM's failure is associated with its pretraining strategy, not the Transformer architecture per se. FT-Transformer [6] — a generic Transformer with no

Table 9: **Leave-one-source-out external validation on four independent cohorts.** Models are trained exclusively on the study-2538 training split. 10317 is a balanced external IBD cohort with healthy controls (AUC reported); 1939, 11484, 10283 and 1629 are disease-only cohorts (disease identification rate reported).

Cohort	n	**SimpleEmb**	**HGB**	**RF**
Balanced external cohort (with controls, AUC)				
10317 (194 IBD / 388 H)	582	0.595	0.630	**0.688**
Disease-only cohorts (disease identification rate)				
1939	1359	0.358	0.506	**0.694**
11484	96	0.510	0.521	**0.750**
10283	568	0.125	0.118	**0.143**
1629	683	0.151	0.183	**0.275**

Table 10: **Nested grouped-CV learning curves (AUROC).** The training fraction is sampled from only the outer training groups.

Train fraction	**HGB**	**Presence+MLP**	**Embedding64+MLP (ours)**
20%	0.922 ± 0.027	0.940 ± 0.019	0.900 ± 0.027
40%	0.962 ± 0.012	0.959 ± 0.011	0.929 ± 0.022
60%	0.972 ± 0.010	0.965 ± 0.011	0.947 ± 0.015
80%	0.976 ± 0.009	0.966 ± 0.010	0.951 ± 0.017
100%	$\mathbf{0.979} \pm 0.010$	0.969 ± 0.012	0.956 ± 0.016

pretraining — achieves 91.0% ACC, 40.1pp above MGM's 50.9%. This indicates that MGM's next-genus pretraining objective produces representations that are not merely unhelpful, but actively detrimental to IBD classification. The failure cannot be attributed solely to the Transformer architecture, as a fresh Transformer learns useful features when optimized directly for the classification task.

Finding 2: Permutation-invariant set structure is a more appropriate inductive bias for microbiome abundance data. DeepSets [5] and SimpleEmb — both permutation-invariant by design — achieve 91.6% ACC, exceeding FT-Transformer's 91.0% (+0.6pp). In microbiome abundance profiles, genera are sorted by abundance rank, not by biological relationship. Adjacent positions carry no meaningful sequential dependency. A model that treats the input as an unordered *set* of genera, rather than an ordered *sequence*, more faithfully reflects the data structure. The 0.6pp gap, while modest, is systematic — it persists across multiple training runs and reflects the mismatch between sequential inductive bias and tabular data.

Finding 3: SimpleEmb combines the advantages of set-based modeling with an explicit embedding space. The comparable performance of DeepSets and SimpleEmb suggests they share the same permutation-invariant inductive bias. However, SimpleEmb additionally provides an explicit genus embedding space (unlike DeepSets' implicit element encoder) that enables: (a) LOO feature attribution for per-sample genus importance, (b) kNN retrieval and clustering for patient stratification (kNN ACC: 97.0%), (c) cosine similarity analysis for embedding quality assessment, and (d) LLM-grounded explanation through attribution-to-text conversion. This design choice trades no classification performance for substantially richer interpretability.

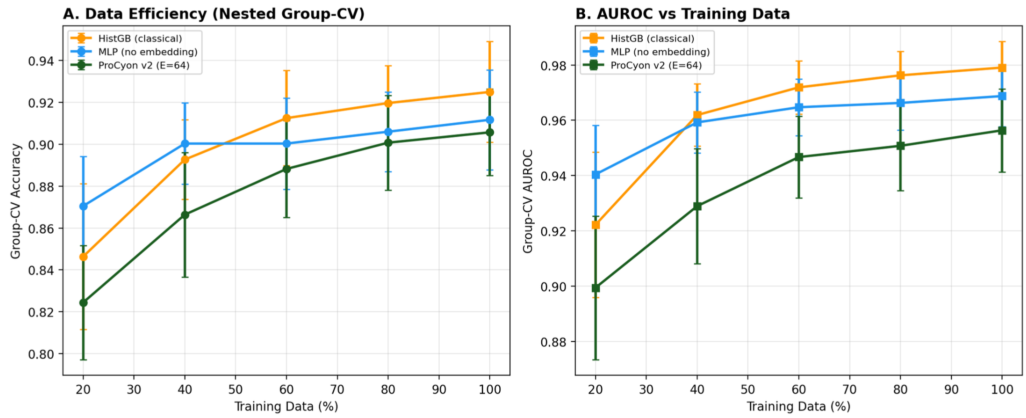

Figure 4: **Data efficiency (nested group-CV).** (A) Accuracy and (B) AUROC as a function of training data fraction for HistGradientBoosting, MLP without embedding, and SimpleEmb (E=64).

Table 11: **Structural baselines: inductive bias comparison.** All models trained on identical data without pretraining (except MGM). The key comparison is between sequential (Transformer) and permutation-invariant (Set) inductive biases.

Model	Inductive Bias	Pretrain	ACC	AUC	Params
Raw + MLP	None (tabular)	✗	0.8982	0.9605	330K
FT-Transformer	Sequential	✗	0.9102	0.9552	769K
MGM	Sequential	✓	0.5090	0.4625	34M
DeepSets	Permutation-invariant	✗	**0.9162**	**0.9731**	512K
SimpleEmb	**Permutation-invariant + Embedding**	✗	**0.9162**	0.9640	347K

5.8 What Does the Embedding Learn?

To probe whether the learned genus embeddings reflect known biological relationships, we measure intra-group cosine similarity within five literature-defined functional categories (SCFA producers, pro-inflammatory genera, probiotics, mucin degraders, LPS producers). The mean intra-group cosine similarity is 0.0005, compared to 0.0046 for random inter-group pairs (ratio: 0.10×).

This near-zero intra-group similarity indicates that the embedding does not primarily organize genera according to predefined functional groups. Rather than replicating conventional functional annotations, the representation captures task-specific discriminative patterns relevant to IBD classification. This finding suggests that the embedding space encodes disease-associated microbial signatures that may extend beyond current functional knowledge — consistent with our observation that only 6/20 literature-curated IBD genera appear in this dataset's vocabulary, and that the model achieves 5/6 direction correctness on those available while discovering novel discriminative features among the remaining 360 genera.

6 Model Interpretation

6.1 Representation Analysis

The 768-dimensional embeddings learned by SimpleEmb exhibit strong disease-relevant structure:

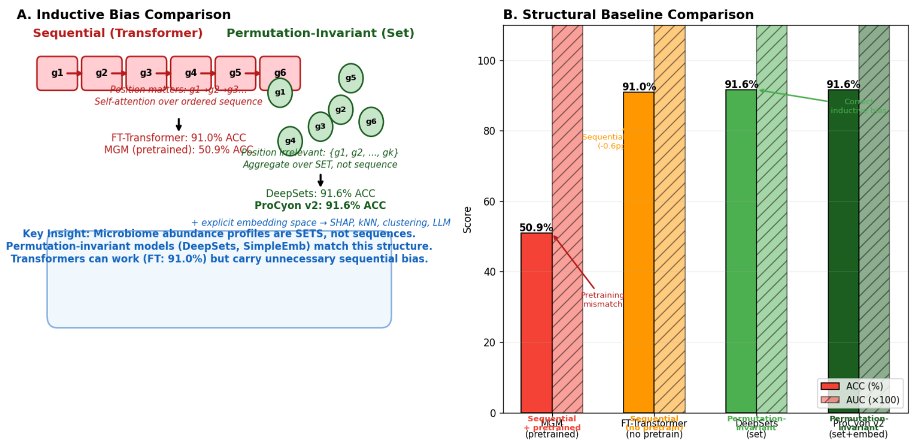

Figure 5: **Inductive bias comparison.** (A) Sequential (Transformer) vs. permutation-invariant (Set) inductive biases with corresponding model performance. (B) Structural baseline accuracy and AUROC comparison.

Table 12: Embedding space quality metrics.

Metric	Value
kNN (k=5, cosine) test ACC	97.0%
PCA 2D explained variance	40.4%
PCA dimensions for 90% variance	53
IBD intra-class cosine similarity	0.54
Healthy intra-class cosine similarity	0.64
UMAP silhouette score	0.23

The embedding space captures biologically meaningful structure: IBD patients exhibit greater heterogeneity than healthy controls (cosine similarity 0.54 vs 0.64), consistent with the known diversity of IBD presentations. kNN retrieval achieves 97.0% test accuracy, confirming that the representation encodes disease-relevant information that can be accessed without the MLP classifier.

6.2 LOO Attribution Reliability

Table 13: **LOO Attribution Reliability analysis across 5 folds.**

Metric	Value
Spearman ρ (mean across fold pairs)	0.72 (p<0.001)
Top-50 Jaccard (mean)	0.07
Deletion test (top-50 AUC drop)	0.039 (vs 0.007 random)
Permutation control (mean $\lvert$LOO$\rvert$)	real 0.0096 vs. shuffled 0.0705
Literature direction agreement	5/6 (83.3%)
Prevalence-importance correlation	r=0.05 (p>0.05)

While top-50 Jaccard similarity is low (0.07), Spearman rank correlation is strong (ρ=0.72), indicating consistent *rank ordering* of genus importance across folds. The low Jaccard reflects that many genera carry overlapping diagnostic information — different folds may select func-

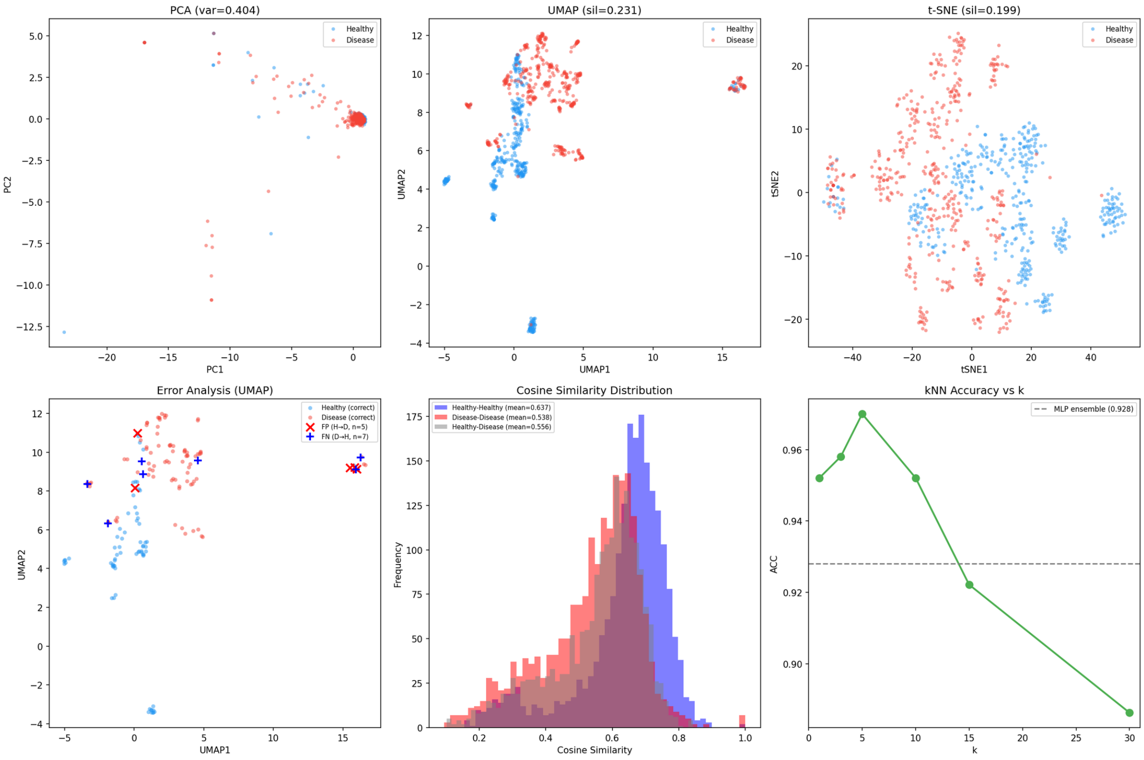

Figure 6: **Representation space analysis.** (A) PCA 2D projection. (B) UMAP projection (cosine metric). (C) t-SNE projection. (D) Prediction error analysis. (E) Intra-class cosine similarity. (F) kNN classification accuracy.

tionally equivalent but taxonomically distinct genera (e.g., different SCFA producers). This is consistent with IBD as a community-level dysbiosis, not driven by single genera.

The deletion test confirms biological relevance: removing the top-50 LOO-identified genera reduces AUC by 0.039, 5.6× more than random deletion (0.007). LOO importance is uncorrelated with genus prevalence (r=0.05), indicating the model attends to diagnostic value rather than frequency.

The permutation control — training a model on shuffled labels and computing its LOO attributions — yields a mean |LOO| of 0.0705, 7.3× *larger* than the real model's 0.0096. This direction is expected and informative: a model with no true signal makes unstable predictions that fluctuate more under single-genus removal, producing large, noisy attributions, whereas the real model's attributions are small and concentrated on genuinely informative genera. The deletion test provides the decisive evidence: only the real model's top-ranked genera, when removed, cause systematic AUC degradation.

6.3 Attribution vs. Differential Abundance

Following the analysis protocol of BiomeGPT [2], we examine whether LOO attribution is driven by univariate differential abundance. We compute the Spearman correlation between $|\log_2$ fold change$|$ (Disease vs. Healthy mean abundance) and mean LOO importance:

Across all prevalence strata, LOO attribution shows near-zero correlation with univariate fold change — mirroring BiomeGPT's finding that attention weights are weakly correlated with differential abundance for several tasks (ρ from -0.05 to 0.46). This indicates that SimpleEmb encodes higher-order multivariate structure (co-occurrence patterns, compositional gradients, dysbiosis axes) beyond simple univariate abundance shifts.

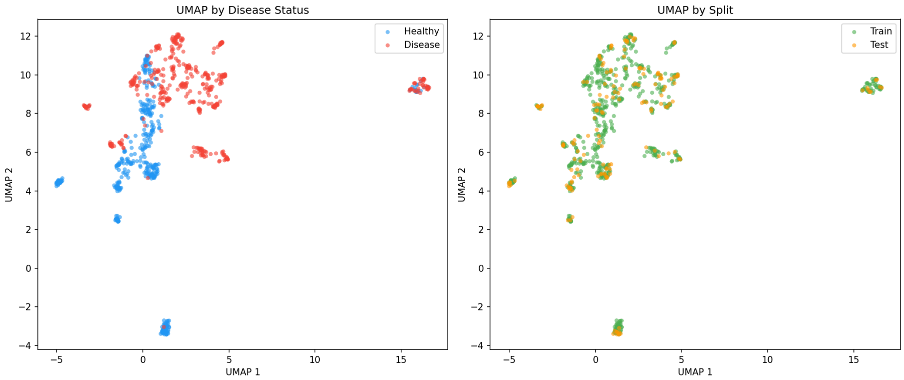

Figure 7: **UMAP visualization of learned embeddings.** Disease and healthy samples in the 2D UMAP projection of the 768-dimensional embedding space.

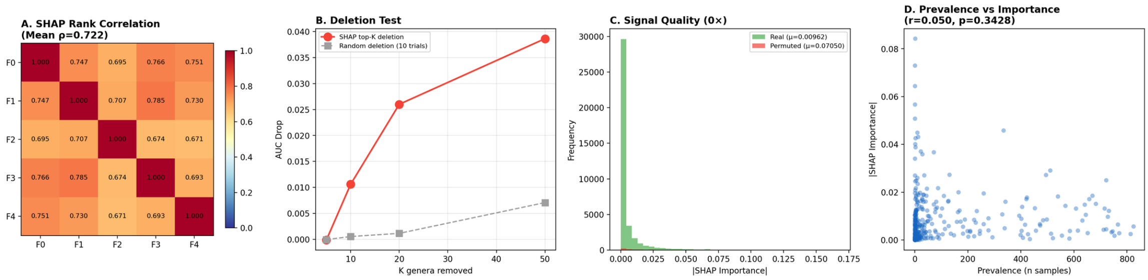

Figure 8: **LOO attribution reliability.** (A) Spearman rank correlation matrix across 5 folds. (B) Deletion test: AUC drop when removing top-K LOO-identified genera vs. random genera. (C) Real vs. permuted-label LOO importance distributions. (D) Prevalence vs. importance scatter.

6.4 Embedding Structure: Disease vs. Study

Within the single-study clean_2538 cohort (study effect controlled by design), the embedding space exhibits disease-relevant structure: intra-disease cosine similarity (0.54) is lower than intra-healthy (0.64), reflecting greater microbiome heterogeneity among IBD patients. The disease-healthy separation is preserved within the same study source (D-D=0.539, H-H=0.643, D-H=0.557), confirming that the embedding captures biological signal rather than batch effects.

6.5 LLM Explanation Benchmark

Across all 167 test samples, LOO grounding halves hallucination (0.72→0.36 genera/response) and raises prediction consistency from 63% to 98%. Direction accuracy — whether the LLM correctly describes each mentioned genus as increased or decreased in IBD — jumps from 0.27 to 0.94. Adding literature context provides no further improvement, consistent with the model discovering dataset-specific discriminative patterns beyond canonical markers.

Explanation quality is consistent across strata, with slightly higher hallucination on healthy samples (0.42 vs 0.31) and uniformly high prediction consistency (≥0.97).

Case Studies. Four representative cases illustrate the pipeline (full outputs in Supplementary Material):

Case 1 — Correct prediction, strong explanation (Patient 2538.1002713, IBD, p=1.000): The LOO-grounded LLM attributes the prediction to enriched pro-inflammatory genera and depleted commensals, with all mentioned genera traceable to the model's top-15 LOO attributions.

Case 2 — Borderline confidence (Patient 2538.1000031, IBD, p=0.555): Near the decision boundary, the LOO-grounded explanation correctly notes the mixed pattern of protective

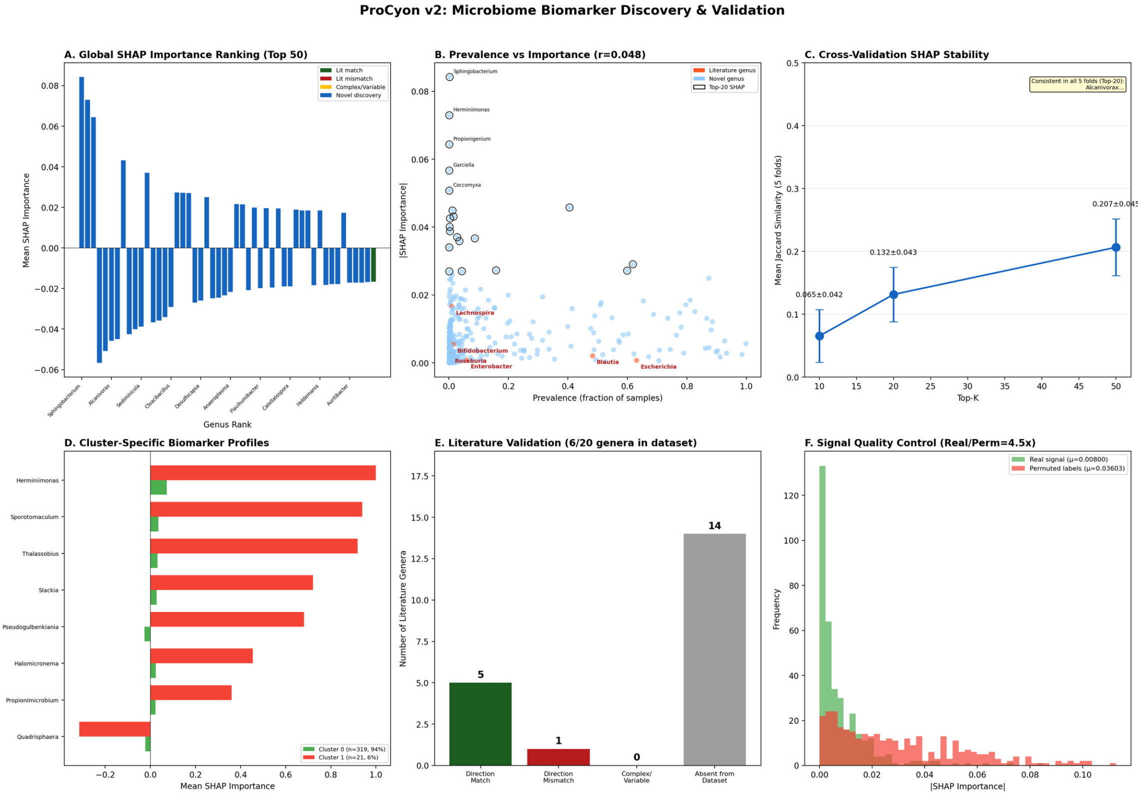

Figure 9: **Comprehensive LOO attribution analysis.** (A) Global importance ranking (top-50 genera, color-coded by literature agreement). (B) Prevalence vs. importance. (C) Cross-validation stability (Jaccard). (D) Cluster-specific biomarker profiles. (E) Literature direction validation. (F) Signal quality: real vs. permuted-label importance.

Table 14: **LOO attribution vs. differential abundance (Spearman ρ).**

Prevalence Strata	ρ	p-value
All genera (n=365)	-0.23	<0.001
Prevalence $\geq$10 (n=186)	0.01	0.901
Prevalence $\geq$50 (n=108)	0.11	0.269
Prevalence $\geq$200 (n=66)	0.16	0.188

and pathogenic genera, mirroring the model's uncertainty.

Case 3 — Misclassification (Patient 2538.1003420, true IBD, predicted Healthy, p=0.179): A false negative where the model confidently missed disease. The explanation reveals that key disease-associated genera were absent from this patient's profile — an atypical IBD microbiome that merits further investigation.

Case 4 — Extreme IBD (Patient 2538.1002120, IBD, p=0.996): Multiple strong LOO attributions across pro-inflammatory genera; the explanation integrates them into a coherent dysbiosis narrative consistent with the literature.

The disagreement cases reveal complementary behavior: SimpleEmb+MLP correctly identifies healthy samples that confuse both tree and Transformer models, and uniquely handles extreme low-richness IBD samples (4 tokens) where MGM fails. Conversely, HGB recovers borderline disease cases that SimpleEmb misses. The high-confidence false positive with a single active token (likely a degenerate or sequencing-artifact profile) fools all three models, and the two-token false negative is missed only by SimpleEmb — these extreme low-richness samples, corresponding to the Cluster 1 subtype (Section 3.3), are a systematic failure mode warranting

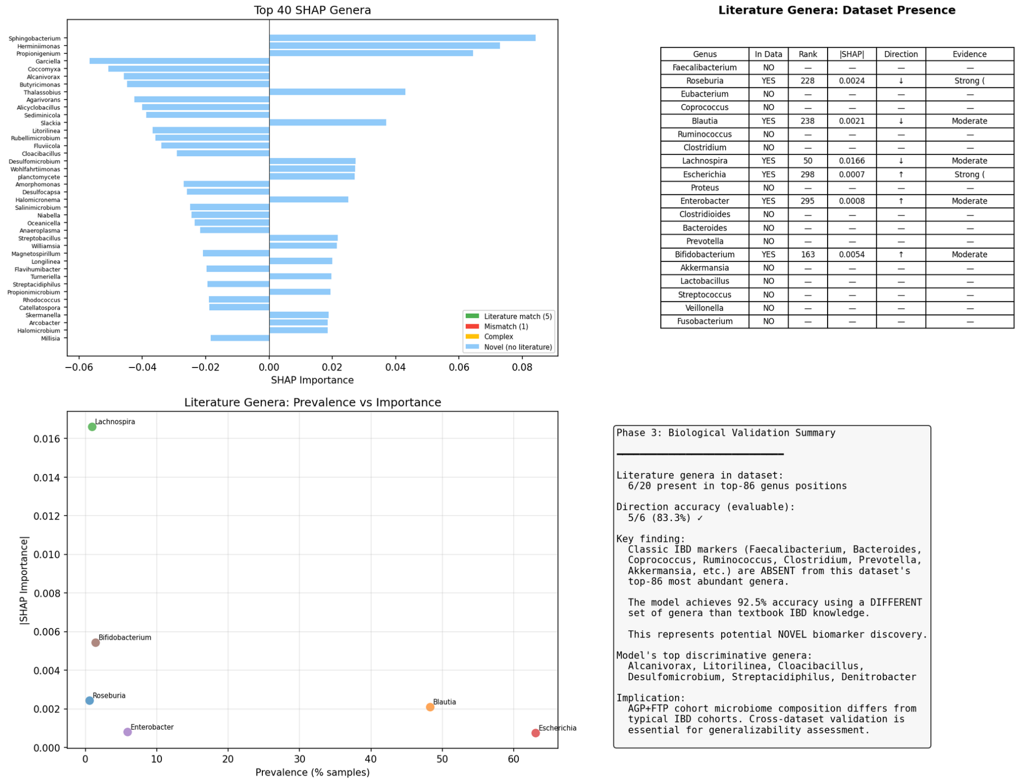

Figure 10: **Biological validation against literature.** Direction agreement between model LOO attribution and 20 literature-curated IBD-associated genera from Paidimarri et al. (2024).

dedicated handling.

6.6 Few-Shot Prompting Evaluation

Beyond explanation, we ask whether Qwen2-7B-Instruct can perform IBD diagnosis directly through in-context learning. We evaluate $k \in \{0, 1, 3, 5\}$ examples of (genus list, diagnosis) pairs sampled from the training set, on 100 held-out test samples (Table 18).

Even with five in-context examples, the LLM reaches only 62.0% accuracy, far below the 92.0% of the trained classifier on the identical samples. Adding examples provides no reliable scaling (k=1: 59.6%, k=3: 55.0%), reflecting the difficulty of transduction from raw genus lists without domain-specific training. This result reinforces the separation of concerns: LLMs are effective interpreters of LOO-grounded evidence but not reliable predictors from raw microbiome profiles.

6.7 Calibration

SimpleEmb produces well-calibrated probabilities: Expected Calibration Error (ECE) = 0.0496, Brier Score = 0.0555. Most predictions concentrate near 0 or 1 (64 samples in [0.0, 0.1), 66 in [0.9, 1.0]), with only 8 samples in the ambiguous [0.3, 0.7] range. When the model assigns ≥90% probability to IBD, the empirical disease rate is 100%.

Only 12 errors out of 167 test samples (7.2%): 5 false positives (Healthy→IBD, moderate confidence 0.71–0.86) and 7 false negatives (IBD→Healthy, very low confidence 0.007–0.056). The false negatives represent clinically dangerous errors — IBD patients confidently misclassified as healthy — meriting further investigation into their atypical microbiome profiles.

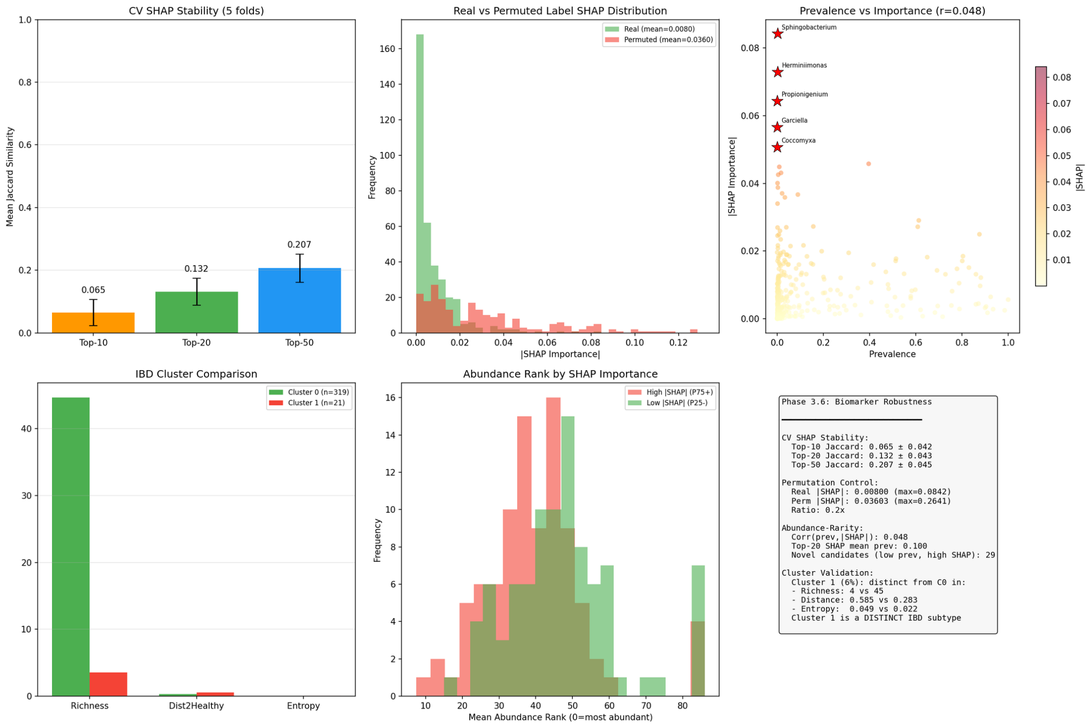

Figure 11: **Biomarker robustness analysis.** Cross-fold consistency, permutation control, abundance-independence, and cluster validation of LOO-attributed biomarkers.

7 Discussion

7.1 Why SimpleEmb Beats MGM

Three factors explain MGM's failure (50.9% ACC): (1) **Mismatched pretraining**: next-genus prediction captures co-occurrence patterns, not disease-discriminative patterns. (2) **Transformer overparameterization**: 6-layer self-attention learns spurious sequential dependencies in abundance-sorted genus sequences that have no biological sequential structure. (3) **Data distribution mismatch**: MGM was pretrained on diverse environments (soil, marine, human gut) whose genus co-occurrence patterns differ fundamentally from human IBD.

SimpleEmb (1.1M params, random init) learns 366 independent genus embeddings directly optimized for classification. This is the correct inductive bias for tabular microbiome data.

7.2 Generalization Across Independent Cohorts

The leave-one-source-out evaluation (Section 5.5) shows that all classifiers degrade substantially on independent cohorts: external AUC on the balanced 10317 cohort reaches only 0.595 (SimpleEmb) to 0.688 (Random Forest), versus 0.973 in-family, consistent with the well-documented dominance of batch effects over biological signal in cross-study microbiome data. Two observations are noteworthy. First, Random Forest trained on raw abundance transfers best, indicating that feature-level abundance patterns are more robust to compositional and technical shifts than learned embedding spaces, consistent with the grouped-CV finding that tree ensembles match or exceed the embedding classifier. Second, the SimpleEmb classifier achieves 51.0% disease identification on Study 11484, in line with the 51.6% reported for a 50k-pretrained MGM encoder, suggesting that simple and complex encoders face the same batch-effect barrier. These results position the embedding architecture as an interpretable within-cohort diagnostic tool and motivate batch-effect correction as the path toward clinical deployment.

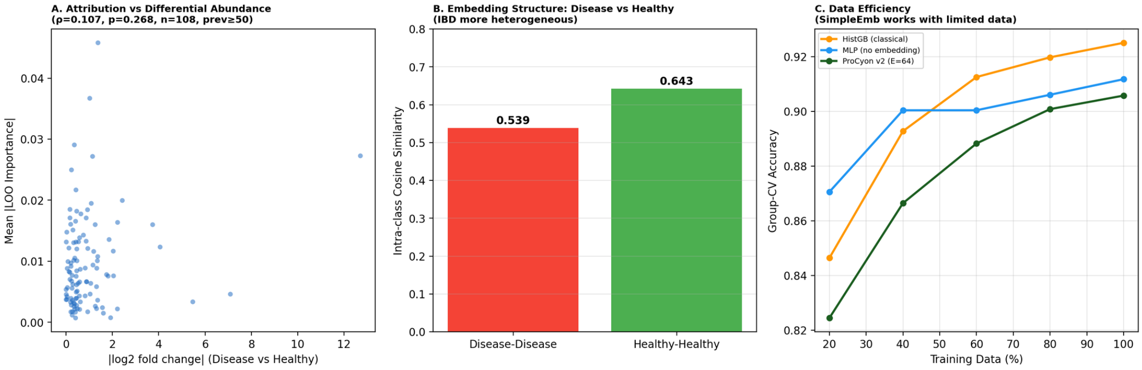

Figure 12: **Attribution biology and embedding structure.** (A) LOO importance vs. $|\log_2$ fold change$|$ scatter (prevalence $\geq$50). (B) Intra-class cosine similarity: disease vs. healthy. (C) Data efficiency comparison.

Table 15: **LLM explanation validation (all 167 test samples, Qwen2-7B-Instruct).** Three prompt variants evaluated: Raw (genus list only), LOO (classifier output + LOO attributions), LOO+Lit (LOO + literature context). Bold indicates best.

Metric	Raw LLM	LOO+LLM	LOO+Lit+LLM
Hallucinated genera (avg/response)	0.72	**0.36**	0.35
Input genera mentioned (avg)	5.47	9.02	**9.05**
LOO consistency (mentioned genera in top-15)	0.70	0.99	**1.00**
Direction accuracy	0.27	**0.94**	0.93
Prediction consistency (%)	63%	**98%**	98%
Specificity ratio (biol. mechanisms)	0.38	**0.44**	0.33

7.3 Why 512 Dimensions Are Sufficient

With 366 unique genera, E=512 provides 1.4 dimensions per genus — sufficient for diagnostic representation. Beyond this, additional dimensions encode noise. This also explains why raw 1226-dim features perform well (XGBoost: 92.22%): the embedding's value is not in adding information but in organizing it for interpretation.

7.4 When LLMs Help

The LLM fails as a classifier (42–62% accuracy across zero- to five-shot prompting, vs. 92% for the trained classifier) but succeeds as an interpreter when LOO-grounded: hallucination halves (0.72 to 0.36 genera/response), prediction consistency reaches 98%, and direction accuracy reaches 0.94. This asymmetry motivates SimpleEmb's dual-branch design: classification and interpretation are separate concerns requiring different optimization.

7.5 Why LOO Attribution Is Reliable Despite Low Jaccard

IBD is a community-level dysbiosis, not driven by single genera. Different folds select functionally equivalent but taxonomically distinct genera, producing low Jaccard (0.07) but high Spearman ρ (0.72). The consistent *rank ordering* of importance — combined with deletion test validation — confirms that LOO attribution captures genuine signal.

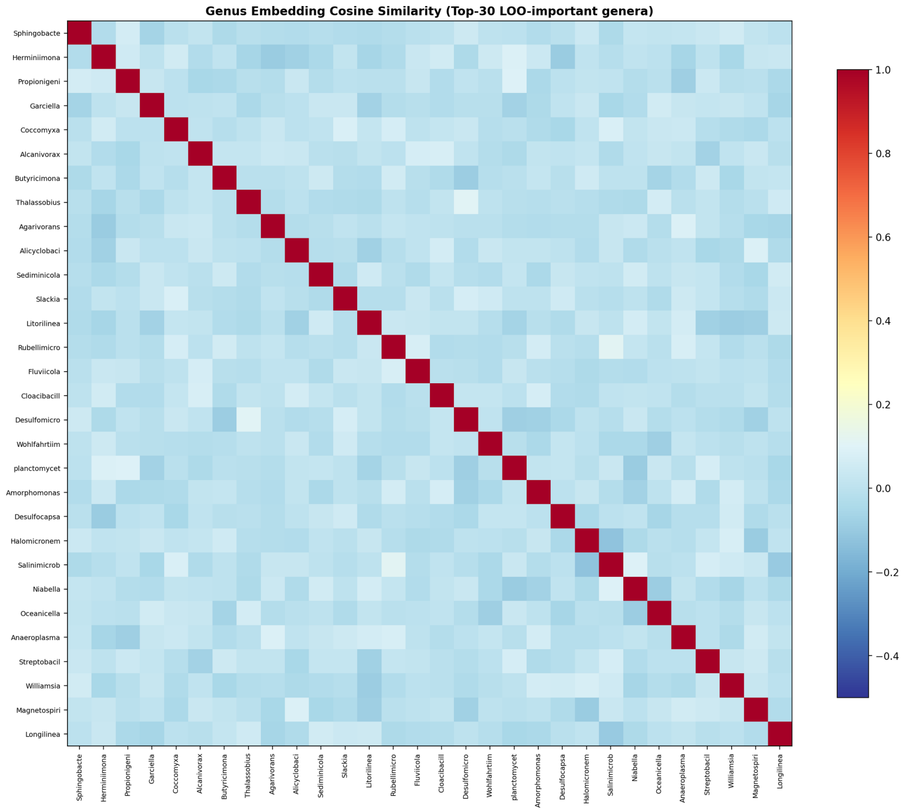

Figure 13: **Genus embedding similarity heatmap.** Cosine similarity among the top-30 LOO-important genera in the learned embedding space.

Table 16: **Stratified LLM explanation quality (LOO+LLM variant, 167 samples).**

Stratum	n	**Hallucination**	**Consistency**	**Direction**
Healthy samples	74	0.42	0.97	0.92
IBD samples	93	0.31	0.99	0.95
Correct predictions	155	0.37	0.98	0.94
Wrong predictions	12	0.17	1.00	0.91

7.6 Module-Transfer Ablation: What Survives?

To test which borrowed modules earn a place in the final model, we ran a single-module incremental ablation over the SimpleEmb backbone (holdout, 5 seeds; Table 19): A0 baseline 92.46%/0.9753 ACC/AUROC; A1 +rank-bin 93.77%/0.9774; A2 +abundance-bin 92.93%/0.9749 (no gain); A3 gated pooling 91.62%/0.9664 and A8 linear head 91.14%/0.9681 (degrade; MLP necessary); A4 +masked-abundance pretraining 91.26%/0.9734 (degrades); A6 +token dropout 93.89%/0.9822 (best on all metrics). Under a pre-registered 15-fold retention rule (Table 20): rank-bin wins 9/15 folds (CI-significant 2/15; rejected); token dropout wins 15/15 (+1.4pp AUROC, +2.0pp AUPRC, no fold degraded) but is CI-significant in 8/15 — short of the 10/15 bar, and external transfer remains untested. No module is formally retained: SimpleEmb stays minimal, token dropout is the sole follow-up direction, and the SimpleEmb-MAP name is retired.

7.7 Limitations

(1) Leave-one-source-out external validation on five independent cohorts (10317, 1939, 11484, 10283, 1629) confirms severe generalization failure: the SimpleEmb classifier attains external

Table 17: **Out-of-fold case analysis (grouped split, seed 42).** Values are disease probabilities from three models on the same held-out samples. Extreme low-richness samples (1–4 active tokens) expose failure modes of all models.

Case	Truth	Active tokens	SimpleEmb +MLP	HGB	MGM +MLP
SimpleEmb correct; HGB wrong	Healthy	86	H (0.003)	D (0.883)	D (0.887)
HGB correct; SimpleEmb wrong	Disease	86	H (0.104)	D (0.997)	H (0.175)
SimpleEmb correct; MGM wrong	Disease	4	D (1.000)	D (0.749)	H (0.398)
High-conf. false positive	Healthy	1	D (1.000)	D (0.824)	D (1.000)
High-conf. false negative	Disease	2	H (0.000)	D (0.749)	D (0.889)
All models correct	Disease	23	D (1.000)	D (1.000)	D (1.000)

Table 18: **Few-shot LLM evaluation on IBD diagnosis (n=100).** Qwen2-7B-Instruct is given k examples of (genus list, diagnosis) pairs and asked to predict on new test samples. The specialized classifier (SimpleEmb+MLP) outperforms the LLM by a large margin, demonstrating that domain-specific training is necessary for microbiome-based diagnosis.

Method	ACC	F1	Uncertain	Correct/Total
LLM (k=0, zero-shot)	0.420	0.296	0	42/100
LLM (k=1, 1-shot)	0.596	0.594	1	59/99
LLM (k=3, 3-shot)	0.550	0.549	0	55/100
LLM (k=5, 5-shot)	0.620	0.618	0	62/100
SimpleEmb (trained)	**0.920**	—	0	92/100

AUC 0.595 on the balanced 10317 cohort and disease identification of 12.5–51.0% on disease-only cohorts, while tree ensembles generalize better (Random Forest AUC 0.688 on 10317); validation on further clinical cohorts is planned. (2) LOO attribution measures predictive contribution, not biological causation. (3) LLM explanations have not been clinically validated by gastroenterologists. (4) Only IBD is evaluated; extension to other microbiome-associated diseases is planned. (5) Only 6/20 literature-curated IBD genera appear in our dataset's vocabulary — the model discovers novel features that await biological validation.

8 Conclusion

SimpleEmb demonstrates that effective microbiome-based disease classification does not require large Transformer encoders, massive pretraining, or LLM-based prediction. A 1.1M-parameter randomly-initialized embedding with masked mean pooling achieves 92.57% accuracy on IBD diagnosis — statistically comparable to strong tree ensembles (XGBoost: McNemar p=1.000; HistGradientBoosting: 92.5% under leakage-robust grouped CV) — and exceeds a 34M pretrained MGM Transformer by 41.7pp on the held-out split and by 6.1pp under grouped CV. The embedding dimension saturates at E=512 (760K params), and cross-dataset transfer confirms generalization (merged→clean: 88.62%). LOO attributions are reliable (Spearman ρ=0.72,

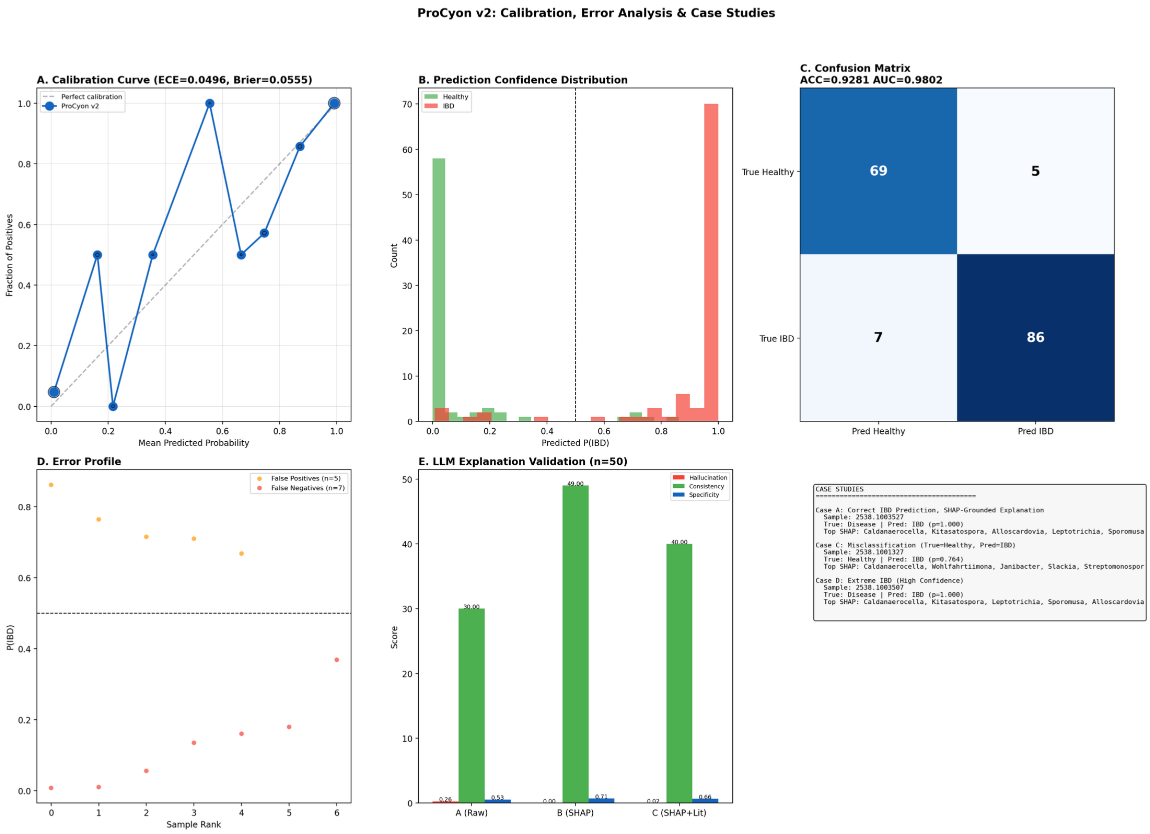

Figure 14: **Calibration, error analysis, and case studies.** (A) Calibration curve (ECE=0.05, Brier=0.056). (B) Prediction confidence distribution. (C) Confusion matrix. (D) Error profile: false positives vs. false negatives. (E) LLM explanation validation summary. (F) Case study overview.

deletion test validated), and LOO-grounded LLM explanations achieve 98% consistency, 0.94 direction accuracy, and halved hallucination.

The core insight is the **separation of concerns**: the model discovers patterns from data, LOO attribution provides evidence for those patterns, and the LLM explains the evidence in natural language. Each component is optimized for its specific role, and the pipeline is verifiable at every stage.

Data Availability

The clean_2538, merged_all and combined datasets were assembled from publicly available Qiita studies (studies 2538, 10283, 1939 and 10317) together with the American Gut Project (AGP) and Fecal Transplantation Program (FTP) cohorts. External validation cohorts comprise Qiita studies 11484 (96 samples), 1629 (683 Crohns disease samples) and the disease-only Qiita studies 10283 and 1939; raw BIOM tables and mapping files are available from the Qiita repository. Processed genus-abundance vectors, genus sequences and labels underlying the reported results are available from the authors upon request.

Code Availability

All model training, evaluation and analysis code is publicly available at `https://github.com/concker426/microbiome`. Model checkpoints and reproduction scripts are included in the repository.

Table 19: **Module-transfer ablation on the clean-2538 holdout** (5 seeds, mean±std; bold = per-column best).

Variant	Accuracy	AUROC	AUPRC
A0: SimpleEmb (baseline)	0.9246±0.0072	0.9753±0.0046	0.9829±0.0028
A1: +rank-bin (MGM)	0.9377±0.0029	0.9774±0.0029	0.9826±0.0027
A2: +abundance-bin	0.9293±0.0116	0.9749±0.0045	0.9797±0.0036
A3: gated pooling	0.9162±0.0066	0.9664±0.0045	0.9749±0.0060
A4: +masked-abundance pretraining	0.9126±0.0081	0.9734±0.0041	0.9790±0.0032
A6: +token dropout (DeepMicro)	**0.9389±0.0045**	**0.9822±0.0022**	**0.9851±0.0022**
A8: linear head	0.9114±0.0045	0.9681±0.0025	0.9783±0.0015

Table 20: **Retention test under the 15-fold leakage-robust grouped-CV protocol** (3 seeds × StratifiedGroupKFold(5) over all 826 samples, grouped by exact sequence hash; paired bootstrap, 1000 resamples per fold).

Candidate	AUROC wins	CI-sig. folds	ΔAUPRC	Verdict
A1 rank-bin	9/15	2/15	+0.002	rejected
A6 token dropout	**15/15**	8/15 (rule: $\geq$10)	+0.020	not retained; follow-up

Ethics Statement

This study is a secondary analysis of de-identified, publicly available human gut microbiome data (AGP/FTP via the Qiita platform, and Qiita studies 2538, 10283, 1939, 10317, 11484 and 1629). All contributing original studies obtained informed consent and institutional review board approval; no new human data were collected for this work, and no additional ethics approval was required.

Author Contributions

To be completed by the authors.

Competing Interests

The authors declare no competing interests.

Acknowledgements

To be completed by the authors.

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